

JESS Thermodynamic Database v8.9

Osmium

24-Jan-23

Reaction No. 5067							
$\text{Os+4_Cl-1(6)} + \text{e-1} = \text{Os+3_Cl-1(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HCl	$\lg K: 14.0 (0.1\text{SD})$	0	ENS	600
2	25	0	Inf. Dilution	$E: 0.85 (2\text{SF})$	4	CRV	22551

Reaction No. 5068							
$\text{OsO}_2(\text{s}) + 2\langle\text{H}_2\text{O}\rangle + 4\langle\text{e-1}\rangle = \text{Os}(\text{s}) + 4\langle\text{OH-1}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -14.5 (3\text{SF})$	1	ELC	
2	25	0	HCl	$\lg K: -10.0 (0.1\text{SD})$	0	ENS	600

Reaction No. 5069							
$\text{Os+8} + \text{Os+6} = 2\langle\text{Os+7}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20:30	1	KOH	$\lg K: 0.9 (0.2\text{SD})$	0	MSP	5465

Reaction No. 5070							
$\text{OsO}_3 + \text{OH-1} = \text{OsO}_3\text{OH-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Unknown	$\lg K: 5.3 (0.05\text{SD})$	0	CSM	233

Reaction No. 5071							
$\text{OsO}_3 + 2\langle\text{OH-1}\rangle = \text{OsO}_3\text{OH-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Unknown	$\lg K: 8.7 (0.05\text{SD})$	0	CSM	233

Reaction No. 5591							
$\text{Thr-1} + \text{Os+4} = \text{Os+4_Thr-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	28	0.02	Unknown	$\lg K: 5.60 (3\text{SF})$	4	MGL	905

Reaction No. 5701							
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2<Met-1> + Os+4 = Os+4_Met-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	28	0.02	Unknown	lgK:6.06(3SF)	6	MGL	905

Reaction No. 5739							
Ser-1 + Os+4 = Os+4_Ser-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	28	0.02	Unknown	lgK:5.60(3SF)	4	MGL	905

Reaction No. 18527							
2<Gly-1> + Os+4 = Os+4_Gly-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	28	0	Cl-1 salt	lgK:5.04(3SF)	4	MGL	1555

Reaction No. 35994							
Os+3 + Thiourea = Os+3_Thiourea							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:4.06(3SF)	4	MPS	4117
2	25		Ethanol:Water 20:80Vol	lgK:4.72(3SF)	4	MPS	4117
3	25		Ethanol:Water 40:60Vol	lgK:4.80(3SF)	4	MPS	4117
4	25		Ethanol:Water 60:40Vol	lgK:4.70(3SF)	4	MPS	4117

Reaction No. 35995							
Os+3 + 2<Thiourea> = Os+3_Thiourea(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:7.54(3SF)	4	MPS	4117
2	25		Ethanol:Water 20:80Vol	lgK:8.87(3SF)	4	MPS	4117
3	25		Ethanol:Water 40:60Vol	lgK:9.08(3SF)	4	MPS	4117
4	25		Ethanol:Water 60:40Vol	lgK:9.06(3SF)	4	MPS	4117

Reaction No. 35996							
Os+3 + 3<Thiourea> = Os+3_Thiourea(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:10.6(3SF)	4	MPS	4117

2	25		Ethanol:Water 20:80Vol	lgK:12.79 (4SF)	4	MPS	4117
3	25		Ethanol:Water 40:60Vol	lgK:13.17 (4SF)	4	MPS	4117
4	25		Ethanol:Water 60:40Vol	lgK:13.14 (4SF)	4	MPS	4117

Reaction No. 35997							
Os+3 + 4<Thiourea> = Os+3_Thiourea(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:13.4 (3SF)	4	MPS	4117
2	25		Ethanol:Water 20:80Vol	lgK:16.55 (4SF)	4	MPS	4117
3	25		Ethanol:Water 40:60Vol	lgK:16.9 (3SF)	4	MPS	4117
4	25		Ethanol:Water 60:40Vol	lgK:17.07 (4SF)	4	MPS	4117

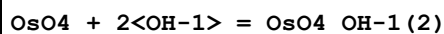
Reaction No. 35998							
Os+3 + 5<Thiourea> = Os+3_Thiourea(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:16.03 (4SF)	4	MPS	4117
2	25		Ethanol:Water 20:80Vol	lgK:19.63 (4SF)	4	MPS	4117
3	25		Ethanol:Water 40:60Vol	lgK:20.59 (4SF)	4	MPS	4117
4	25		Ethanol:Water 60:40Vol	lgK:20.82 (4SF)	4	MPS	4117

Reaction No. 35999							
Os+3 + 6<Thiourea> = Os+3_Thiourea(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:18.4 (3SF)	4	MPS	4117
2	25		Ethanol:Water 20:80Vol	lgK:23.48 (4SF)	4	MPS	4117
3	25		Ethanol:Water 40:60Vol	lgK:23.99 (4SF)	4	MPS	4117
4	25		Ethanol:Water 60:40Vol	lgK:24.31 (4SF)	4	MPS	4117

Reaction No. 54699							
OsO4 + H2O = OsO4_OH-1 + H+1							

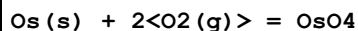
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.3(0.1SD)	4	MDS	5103

Reaction No. 54700



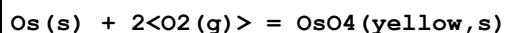
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:1.1(0.05SD)	7	CSM	8

Reaction No. 54731



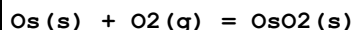
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-72.(1.SD)	3	ENS	869

Reaction No. 54732



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:53.(0.5SD)	3	ENS	802
2	25	0	Inf. Dilution	lgK:53.43(4SF)	4	ENS	6422
3	27	0	Inf. Dilution	lgK:53.00(4SF)	4	ENS	6422
4	31	0	Inf. Dilution	lgK:52.10(4SF)	4	ENS	6422
5	25	0	Inf. Dilution	dG:-72.88(4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dH:-94.(0.5SD)	4	ENS	802
7	25	0	Inf. Dilution	dH:-94.19(4SF)	4	MTD	6422
8	27	0	Inf. Dilution	dH:-94.16(4SF)	4	MTD	6422
9	31	0	Inf. Dilution	dH:-94.10(4SF)	4	MTD	6422

Reaction No. 54733



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.97(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:41.65(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:28.82(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:21.15(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:16.06(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-46.(1.SD)	0	ENS	869
7	25	0	Inf. Dilution	dG:-57.25(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-70.5(3SF)	4	MTD	6422

9	127	0	Inf. Dilution	dH:-70.3(3SF)	4	MTD	6422
10	227	0	Inf. Dilution	dH:-70.1(3SF)	4	MTD	6422
11	327	0	Inf. Dilution	dH:-69.7(3SF)	4	MTD	6422

Reaction No. 54734							
Os(s) + 2<O2(g)> = OsO4(white,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:53.(0.5SD)	3	ENS	802
2	25	0	Inf. Dilution	dH:-92.(0.5SD)	4	ENS	802

Reaction No. 54735							
Os(s) + 2.5<O2(g)> + H2(g) = H+1_OsO4_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:95.(0.9SD)	0	ENS	869

Reaction No. 54736							
Os(s) + 2.5<O2(g)> + 2<e-1> = OsO5-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-93.(1.SD)	3	ENS	869

Reaction No. 54737							
Os(s) + 4<OH-1> = OsO2.2H2O(s) + 4<e-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.12(0.02SD)	4	ENS	869

Reaction No. 57792							
Os(s) = Os(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:177.(1.SD)	4	ENS	869
2	25	0	Inf. Dilution	dH:188.(1.SD)	4	ENS	869

Reaction No. 57793							
Os(s) + 2<O2(g)> + 2<H2(g)> = OsO2.2H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:118.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-161.(1.SD)	0	ENS	869

Reaction No. 57794							
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Os(s) + 2<O2(g)> = OsO4(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-70.(1.SD)	4	ENS	869
2	25	0	Inf. Dilution	dH:-81.(1.SD)	4	ENS	869

Reaction No. 57795							
Os(s) + H2(g) + 2.5<O2(g)> = H+1(2)_OsO5-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-129.(1.SD)	4	ENS	869

Reaction No. 57796							
Os(s) + 0.5<H2(g)> + 2.5<O2(g)> = H+1_OsO5-2 - e-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-113.(1.SD)	4	ENS	869

Reaction No. 59562							
Asn-1 + Os+4 = Os+4_Asn-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	28	0.02	Unknown	lgK:5.17(3SF)	3	MGL	905

Reaction No. 59994							
2<Ala-1> + Os+4 = Os+4_Ala-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.97(3SF)	0	MGL	2392

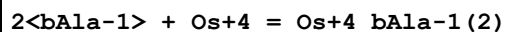
Reaction No. 60002							
2<Val-1> + Os+4 = Os+4_Val-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	28			lgK:7.77(3SF)	3	MGL	2392

Reaction No. 60021							
2<Leu-1> + Os+4 = Os+4_Leu-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	28			lgK:5.82(3SF)	3	MGL	2392

Reaction No. 60027							
2<Ile-1> + Os+4 = Os+4_Ile-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

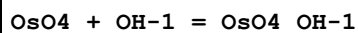
1	28			lgK:5.72 (3SF)	0	MGL	2392
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Reaction No. 60077



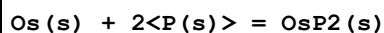
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	28			lgK:6.12 (3SF)	3	MGL	2392

Reaction No. 60510



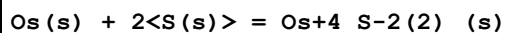
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	Unknown	lgK:1.6 (0.1SD)	5	CRV	2556
2	25	1	Unknown	lgK:1.3 (2SF)	5	CRV	2556

Reaction No. 67129



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.97 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:24.81 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:18.12 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:14.09 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:11.40 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-34.07 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-36.4 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-36.8 (3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-36.9 (3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-37.0 (3SF)	4	MTD	6422

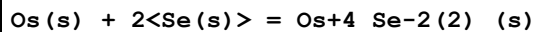
Reaction No. 67130



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.66 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:23.50 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:17.04 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:13.01 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:10.27 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-32.28 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-35.3 (3SF)	4	MTD	6422

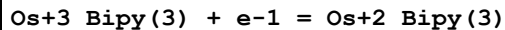
8	127	0	Inf. Dilution	dH:-36.5 (3SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-37.3 (3SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-37.9 (3SF)	0	MTD	6422

Reaction No. 67131



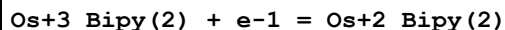
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.18 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:19.05 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:13.80 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:10.60 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:8.224 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-26.17 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-28.7 (3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-29.0 (3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-32.2 (3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-32.9 (3SF)	4	MTD	6422

Reaction No. 71054



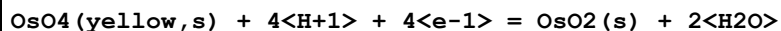
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.80 (2SF)	2	CRV	6330
2	25	0	Inf. Dilution	E:0.885 (3SF)	2	CRV	22551

Reaction No. 71055



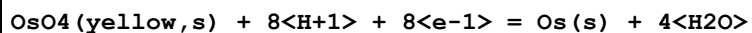
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.81 (2SF)	5	CRV	6330

Reaction No. 71056



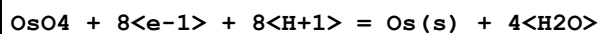
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1	25	0	Inf. Dilution	lgK:72.0 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:1.02 (3SF)	0	CRV	6330

Reaction No. 71057



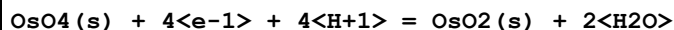
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.838 (3SF)	5	CRV	6330

Reaction No. 74286



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:113.4 (4SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.84 (2SF)	0	CRV	22551

Reaction No. 74287



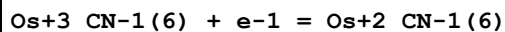
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.005 (4SF)	4	CRV	22551

Reaction No. 74288



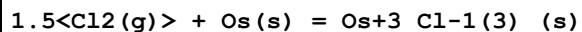
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.5 (3SF)	1	ELC	
2	25	0	Inf. Dilution	E:0.687 (3SF)	0	CRV	22551

Reaction No. 74289



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.634 (3SF)	4	CRV	22551

Reaction No. 75620



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-190.4 (4SF) kJ	3	CRV	6330

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination

6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review
CSM	Smith & Martell Vols 1-6 (crit.)
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
MDS	Distribution between two phases
MGL	Glass electrode
MPS	Potentiometry and Spectrophotometry
MSP	Spectroscopy
MTD	Temperature difference

References

Reference No.: 8(76SmM)

Critical Stability Constants, Vol. 4, 1976 Smith RM; Martell AE; Plenum, New York

Reference No.: 233(89SmM)

Critical Stability Constants, Vol. 6, 1989 Smith RM; Martell AE; Plenum, New York

Reference No.: 600(52Lat)

The Oxidation States of the Elements, 2nd Edn, 1952 Latimer WM; and their Potentials in Aqueous Solutions; Prentice-Hall Inc., New York, USA

Reference No.: 802(90Wea)

CRC Handbook of Chemistry and Physics, 70th Edn., 1990 Weast RC; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio

Reference No.: 869(68GoH)

Chem. Rev., 1968, 68, 229 - 252; Goldberg RN; Hepler LG; Thermochemistry and Oxidation Potentials of the Platinum Group Metals and their Compounds

Reference No.: 905(95Ber)

Pure Appl. Chem., 1995, 67, 1117 - 143; Berthon G; Critical Review of Formation Constants of Selected Amino Acids with Polar Side Chains

Reference No.: 1555(91KSG)

Pure Appl. Chem., 1991, 63, 597 - 638; Kiss T; Sovago I; Gergely A; Critical Survey of Stability Constants of Complexes of Glycine

Reference No.: 2392(93SKG)

Pure Appl. Chem., 1993, 65, 1029 - 1080; Sovago I; Kiss T; Gergely A; Critical Survey of the Stability Constants of Complexes of Aliphatic Amino Acids

Reference No.: 2556(93MSM)

NIST Critical Stability Constants of Metal Complexes Database, 1993 Martell AE; Smith RM; Motekaitis RJ; Version 1.0; National Institute of Standards and Technology, Gaithersburg, USA

Reference No.: 4117(83SeS)

Zh. Neorg. Khim., 1983, 28, 3090 - 3094; Sereda IP; Stadnik AM; Study of the Stepwise Complexing of Osmium with Thiourea

Reference No.: 5103(72Lee)

J. Inorg. Nucl. Chem., 1972, 34, 375 - 378; Lee DA; Extraction of Osmium Tetroxide and Evidence for Anionic Osmium(VIII) Species

Reference No.: 5465(68BHC)

C. R. Acad. Sci., 1968, 66, 496 - 499; Beaufils J-P; Hellin M; Coussemant F; Disproportionation Equilibrium in Septivalent Osmium

Reference No.: 6330(95LiF)

CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 6422(95BaP)

Thermochemical Data of Pure Substances, 3rd Edn., 1995 Barin I; Platzki G; Weinheim, Germany

Reference No.: 22551(06Inz)

Encyclopedia of Electrochemistry, Vol. 7a, 2006, 17 - 75; Inzelt G; Standard, Formal and Other Characteristic Potentials of Selected Electrode Reactions